



Scoreboard

Hard Disk Drives

BRAND	Hitachi	Hitachi	Hitachi	Maxtor	Maxtor	Samsung
Model (Series)	Deskstar	Deskstar	Deskstar	DiamondMax 10	Maxline III	SpinPoint P
Model Number	HDS728080PLA380	HDT722525DLA380	HDS725050KLA360	6B300S0		HD080HJ
Features	16.75	16.75	16.75	9.82	9.82	10.10
Capacity (GB)	80	250	500	300	300	80
Formatted Capacity (GB)	76.69	232.88	465.76	279.48	279.48	74.53
Spindle Speed (RPM)	7200	7200	7200	7200	7200	7200
Interface Type (ATA133/SATA)	SATA - II	SATA - II	SATA - II	SATA - I	SATA - I	SATA - II
Interface Speed	3.0 Gbit/s	3.0 Gbits/s	3.0 Gbits/s	1.5 Gbits/s	1.5 Gbits/s	3.0 Gbits/s
Data Buffer Size (MB)	8	8	8	16	16	8
Power Connector	Molex + SATA	Molex + SATA	Molex + SATA	SATA	SATA	SATA
Additional Features	Backward compatible to SATA 1.5Gbits/s, S.M.A.R.T., NCQ, ECC & CRC protection	Backward compatible to SATA 1.5Gbits/s, S.M.A.R.T., NCQ, ECC & CRC protection	Backward compatible to SATA 1.5Gbits/s, S.M.A.R.T., NCQ, ECC & CRC protection	NCQ, shock protection system(Maxtor), hot plug support	NCQ, hot-plug and asynchronous signal recovery	NCQ, SMART compliant, Hot-plug & hot-swap capable
Operating Shock Resistance (in G)	55	55	55	60	60	63
Performance	33.35	36.86	38.22	36.39	35.41	31.90
Synthetic tests						
HD Tach						
Write Speeds						
Min/Max (in MB/s)	10.7/65.3	18.8 / 47	12.3/95	16.1/40.9	11.1/53	18.3/32.8
Average (in MB/s)	28	30.4	29.3	32.4	30.7	25.7
Read Speeds						
Min/Max (in MB/s)	33.2/63.9	29.6 / 72.6	32/67.2	36.2/126.1	36.9/70.8	27.6/65.7
Average (in MB/s)	50.3	56.3	52	56.9	36.9	49.7
Random Access Time (in msec.) *	13.3	12.9	12.9	16.9	17	13.2
Read Burst Speed (MBps)	225	224.9	225.2	138.5	139.1	180.8
CPU Utilisation (%) *	7	5	7	4	5	7
SiSoft Sandra 2005 Pro						
Drive Index (MB/s)	50	56	56	55	51	52
Sequential Read/Write (MBps)	58/58	64/63	62/62	63/62	62/63	59/60
Random Read/Write (MBps)	41/30	46/36	46/50	40/48	32/48	41/44
Access Time (ms) *	7	6	6	9	15	8
Real world tests						
Inter Drive Test (Two drives) *						
Assorted Data (700MB)	31/33	29/33	28/33	30/31	33/32	30/30
Write/Read (in secs)						
Sequential Data (700MB)	18/19	19/19	19/20	19/20	21/22	20/21
Write/Read (in secs)						
Inter Drive Test (Using Ram Drive)*						
Assorted Data (700MB)	22/14	18/13	15/13	13/13	13/14	16/14
Write/Read (in secs)						
Sequential Data (700MB)	13/14	12/12	13/12	11/11	11/12	13/13
Write/Read (in secs)						
Intra Drive Test (With in same Drive)*						
700MB Assorted/Sequential (in secs)	43/37	34/31	29/25	35/26	32/23	50/34
Photoshop Image Load Time *						
200 MB (secs)	4	4	4	5	4	5
550 MB (secs)	51	45	42	45	43	56
800 MB (secs)	78	70	65	71	69	100
1 GB (secs)	116	102	99	106	99	140
1.4 GB (secs)	167	151	122	154	141	213
Warranty (No. Of Years)	5	5	5	3	5	3
Price Index	10.66	15.11	11.33	17.00	16.00	10.99
Cost/GB (Rs) *	45.64	32.21	42.94	28.62	30.41	44.28
Price (Rs)	3500	7,500	20,000	8000	8500	3300
Overall Score (Out of 100)	70.77	78.72	76.31	69.21	71.22	58.99

* Lower scores indicate better performance



+ Molex and SATA connectors
- High CPU use



+ Good performance,
- None



+ Excellent performance,
- High CPU use



+ Great value for money, good performance
- None



+ Good value for money
- None



+ None
- High CPU use